iLearn Practice Exam

10BR · Chapter / Topic 2

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 2: Data Transmission

Name: _____

Class: _____

Date: _____

Time allowed: 40 minutes

TOTAL MARKS: 33

Marks obtained: ..... / 33

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 33.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 5 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, mark your work against the official Cambridge mark scheme (see the source list on the last page).
· Only 33 marks of past-paper questions are available for this chapter, so the recommended time is 40 minutes (1 min 1

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1 Data can be transmitted from one device to another.

(a) Tick (✓) **one** box to show which of the terms is **not** a method for transmitting data.

A serial ☐

B simplex ☐

C parallel ☐

D parity ☐

[1]

(b) Data is broken down into smaller units to be transmitted from one device to another.

Give the name of the unit that data is broken down into.

..... [1]

(c) Data is often encrypted when it is transmitted from one device to another.

(i) Explain how data is encrypted using symmetric encryption.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) Give the purpose of encryption.

.....

..... [1]

2 The table contains statements about error detection methods.

Complete the table by giving the correct error detection method for each statement.

error detection method	statement
.....	An odd or even process can be used.
.....	A value is calculated from the data, using an algorithm. This happens before and after the data is transmitted.
.....	A copy of the data is sent back to the sender by the receiver.
.....	Acknowledgement and timeout are used.
.....	A value is appended to data that has been calculated using the data. This value is checked on data entry.

[5]

3 An optical mouse is connected to a computer using a universal serial bus (USB) cable.

(a) The USB cable transmits data using the serial half-duplex data transmission method.

Describe how the serial half-duplex data transmission method transmits the data.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) One benefit of using the USB cable to connect the optical mouse to the computer is that bits of data are less likely to arrive out of order.

Give **two** other benefits of using the USB cable to connect the optical mouse to the computer.

1 .....

.....

2 .....

.....

[2]

4 A video streaming service uses hexadecimal to display error codes when problems occur with their videos.

(a) The videos are stored on a web server.

The data for the videos is sent from the web server to a user's computer using packet switching.

(i) Each packet of data is divided into three parts. One part is the payload.

Give the names of the **two** other parts of a packet of data.

1 .....

2 .....

[2]

(ii) Explain the role of a router in packet switching.

.....

.....

.....

.....

.....

..... [3]

5 An odd parity byte check is used to check for errors in data that is transmitted from a computer to a printer.

(a) Three bytes of data need to be transmitted.

Give the correct parity bit that would be added to each byte of data for an odd parity byte check.

parity bit	byte of data
.....	0 1 0 1 1 1 0
.....	1 1 1 0 1 1 1
.....	1 0 0 0 1 0 0

[3]

(b) There is an error in the data after transmission. This error is **not** detected by the parity byte check.

(i) Explain why the error may **not** be detected by a parity byte check.

.....

.....

.....

..... [2]

(ii) Give **two** other error detection methods that could be used.

1 .....

2 .....

[2]

(c) The data that is transmitted is encrypted.

(i) Give **one** method of encryption that could be used.

..... [1]

(ii) State what is meant by an encryption key.

.....

..... [1]

(iii) State the name given to the encrypted data.

..... [1]

END OF QUESTION PAPER

Source of each question (for marking)

Find the matching official Cambridge mark scheme for the series/paper shown, to mark your answers.

Q	Past paper	Original question	Marks
1	May/June 2024 Paper 12	Q2 (a),(b),(c)	7
2	May/June 2024 Paper 12	Q6	5
3	May/June 2026 Paper 11	Q4 (b),(c)	6
4	May/June 2026 Paper 12	Q3 (e)	5
5	May/June 2026 Paper 12	Q4 (a),(b),(c)	10

Total: 33 marks